

# Contents of the shipping box



Vangskroken 2  
N-1351 RUD  
Norway  
Tel: +47 6717 4500  
Fax: +47 6713 6770  
inquiry@nortek-as.com  
www.nortek.no

Nortek order no:

40112-7-7-80

Type of system:

SIGNATURE 100

## Instrument type:

- |                                            |                                         |                                            |                                       |                                        |
|--------------------------------------------|-----------------------------------------|--------------------------------------------|---------------------------------------|----------------------------------------|
| <input type="checkbox"/> Aquadopp          | <input type="checkbox"/> Aquadopp 6000m | <input type="checkbox"/> Vector            | <input type="checkbox"/> Signature55  | <input type="checkbox"/> Signature1000 |
| <input type="checkbox"/> Aquadopp profiler | <input type="checkbox"/> AWAC           | <input type="checkbox"/> Vectrino          | <input type="checkbox"/> Signature250 | <input type="checkbox"/> NortekDVL     |
| <input type="checkbox"/> Aquadopp 3000m    | <input type="checkbox"/> VM AWAC        | <input type="checkbox"/> Vectrino profiler | <input type="checkbox"/> Signature500 |                                        |

Software version:

SIGNATURE  
DEPLOYMENT

Firmware version:

7.4.4616.2206\_7/164

Other:

## Cable:

☒ 10m length

☐ Other:

## Connector:

☐ 8-pin Inline

☐ 12-pin UW

☒ 6-pin Inline

☐ 7-pin Souriau

☐ Other:

## Communication:

☐ RS232 ☒ Ethernet

☐ RS422 ☐ Other:

## Options:

☐ Analog input ☐ Synch

☐ Analog output ☐ Other:

## Battery cannister:

- ☐ Paradopp battery cannister
- ☐ Single battery aluminium cannister
- ☐ Double battery aluminium cannister

## Battery cables:

- ☐ 2pin Inline-2pin
- ☐ 8pin Inline-2pin
- ☐ 8pin rectangular-2pin

## Accessories:

- ☒ Toolkit
- ☐ Quick guide
- ☐ Warranty card
- ☒ Final test checklist
- ☐ Seeding material
- ☐ USB to serial converter RS232
- ☐ Altronix AL310 USB driver
- ☐ Recorder kit/ProLog
- ☐ Battery harness for 2 batteries

## Batteries:

- |                                         |                    |                          |
|-----------------------------------------|--------------------|--------------------------|
| <input type="checkbox"/> Alkaline 50Wh  | 13.5V              | <input type="checkbox"/> |
| <input type="checkbox"/> Alkaline 100Wh | 13.5V              | <input type="checkbox"/> |
| <input type="checkbox"/> Alkaline 540Wh | 13.5V              | <input type="checkbox"/> |
| <input type="checkbox"/> Alkaline 540Wh | 18V                | <input type="checkbox"/> |
| <input type="checkbox"/> Alkaline 90Wh  | 15V(Signature1000) | <input type="checkbox"/> |
| <input type="checkbox"/> Alkaline 180Wh | 18V(Signature500)  | <input type="checkbox"/> |

## Extra set:

## AC/DC Power supply

- ☐ 15V standard ☐ 48V Signature55 Online ☐ 24V Vectrino
- Plug
- ☒ 24V DC/DC & Signature ☐ EU ☐ UK ☒ US

Other:

MIDAS SOFTWARE

Date:

23.03.2018

Responsible:

Ronny Johansen



# Final test checklist AD2CP



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Order number:  
  
40112-1-1-80  
(RMA 5174)

Name: Signature 100  
Instrument serial number: 100767  
Frequency: 100 KHz Main board: AD2CP 726  
Firmware versions: 1.4.4616.2206-7/164

Label checked ☒ OK  
Dock test ☒ OK  
Baudrate 115200 ☒ OK

Comments:

Tilt check  
☒ Pitch up  
☒ Roll up  
☒ Status bit  
☒ Pitch down  
☒ Roll down  
pitch & roll within +/- 0.2 °

Clock  
☒ Set clock  
Heading  
☒ Up  
☒ Down  
tolerance: +/- 2 °

Pressure  
Psenortemp ☒ OK  
tolerance: +/- 0.1 % of  
1500 m

Temperature  
☒ OK  
tolerance: +/- 0.1 °

Beam check

| Correct order                                 | Beam Imp | Noise floor | Amplitude in tank | Range                                  |
|-----------------------------------------------|----------|-------------|-------------------|----------------------------------------|
| Beam 1 <input checked="" type="checkbox"/> OK | 77 Ω     | 27 dB       | >80 dB            | <input checked="" type="checkbox"/> OK |
| Beam 2 <input checked="" type="checkbox"/> OK | 76 Ω     | 27 dB       | >80 dB            | <input checked="" type="checkbox"/> OK |
| Beam 3 <input checked="" type="checkbox"/> OK | 77 Ω     | 27 dB       | >80 dB            | <input checked="" type="checkbox"/> OK |
| Beam 4 <input checked="" type="checkbox"/> OK | 77 Ω     | 27 dB       | >80 dB            | <input checked="" type="checkbox"/> OK |
| Beam 5 <input checked="" type="checkbox"/> OK | 593 Ω    | 27 dB       | >80 dB            | <input checked="" type="checkbox"/> OK |

Velocity direction To heavy to Verify.  
XYZ coordinate system Matrix correct.  
ENU coordinate system  
X ☒ OK E ☒ OK  
Y ☒ OK N ☒ OK  
Z ☒ OK U ☒ OK  
Z2 ☒ OK U2 ☒ OK

Head file  
☒ Headfile checked  
☒ Saved as read only

Serial communication  
☒ RS422  
☒ RS232

Recorder erased  
☒ OK  
Rec size: 128GB

Ethernet  
MAC address: 8C:68:78:00:02:D6  
Static IP address:  
Set host name: 100767

DHCP enabled ☒  
FTP OK ☒

Licenses

|                                                    |                                                  |                                                    |                                         |                                                                |
|----------------------------------------------------|--------------------------------------------------|----------------------------------------------------|-----------------------------------------|----------------------------------------------------------------|
| Averaging mode <input checked="" type="checkbox"/> | Wave mode <input type="checkbox"/>               | Dual frequency high <input type="checkbox"/>       | 256GB recorder <input type="checkbox"/> | Calibration license erased <input checked="" type="checkbox"/> |
| Burst Five beams <input type="checkbox"/>          | Echo Sounder <input checked="" type="checkbox"/> | 16GB recorder <input type="checkbox"/>             |                                         | Production license erased <input checked="" type="checkbox"/>  |
| Bottom track <input type="checkbox"/>              | Ice Measurement <input type="checkbox"/>         | 64GB recorder <input type="checkbox"/>             |                                         | Default configuration set <input checked="" type="checkbox"/>  |
| High Resolution <input type="checkbox"/>           | Dual frequency low <input type="checkbox"/>      | 128GB recorder <input checked="" type="checkbox"/> |                                         |                                                                |

Cable/Harness  
Communication ☒ Cable  
Communication ☒ Harness  
Battery ☒ Battery ☒

External sensors

Power down  
☒ OK

Date  
Day 23 Month 03 Year 18  
Signature





# Certificate of Calibrations and Tests

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## Instrument Information

|                          |                      |
|--------------------------|----------------------|
| Customer Reference No.   | 40112-1-1-80         |
| Instrument Type          | Signature100         |
| Instrument Frequency     | 100 kHz              |
| Instrument S/N           | 100767               |
| Head S/N                 | D-0767               |
| Interface Board S/N      | 726                  |
| Interface Board Mfr. S/N | 4MO0283710019        |
| Digital Board Mfr. S/N   | 4MO0356150024        |
| Analog Board Mfr. S/N    | 4MO036890013         |
| Analog Board #2 Mfr. S/N | 4MO0317350004        |
| Sensor Board Mfr. S/N    | 4MO0319780047        |
| Interface Board Rev.     | G-1(High-Power)      |
| Digital Board Rev.       | I-3                  |
| Analog Board Rev.        | A-1(High-Power Echo) |
| Analog Board #2 Rev.     |                      |
| Sensor Board Rev.        | I-0                  |

## Calibrations and tests performed

|                  |        |
|------------------|--------|
| Tilt and Compass | Passed |
| Pressure         | Passed |

All the tested values are within Nortek AS specifications

March 22, 2018

Date

  
Reviewed and approved (sign.)





# Tilt and Compass Report

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## Details

|                 |                       |
|-----------------|-----------------------|
| Instrument Type | Signature100          |
| Instrument S/N  | 100767                |
| Date            | March 20, 2018        |
| Operator        | Asle Martinsen        |
| Location        | Nortek Factory Norway |
| Result          | Passed                |

## Description

Calibration and verification is performed in a two axis automated jig. Continuous and fixed-point measurements are collected to calibrate and verify the sensor.

Criteria of acceptance for tilt sensor is  $\pm 0.2^\circ$ .

Criteria of acceptance for compass sensor is  $\pm 2^\circ$ .

Reference: Digital Protractor Series 950 Pro 3600. Accuracy  $\pm 0.05^\circ$ .

## Tilt Verification Results

| Reference ( $^\circ$ ) | Diff. Up           |                   | Diff. Down         |                   |
|------------------------|--------------------|-------------------|--------------------|-------------------|
|                        | Pitch ( $^\circ$ ) | Roll ( $^\circ$ ) | Pitch ( $^\circ$ ) | Roll ( $^\circ$ ) |
| -30.00                 | -0.12              | -0.04             | -0.12              | -0.07             |
| -15.00                 | -0.06              | 0.05              | 0.04               | 0.10              |
| 0.00                   | 0.11               | 0.02              | 0.12               | -0.01             |
| 15.00                  | -0.13              | -0.06             | -0.13              | -0.01             |
| 30.00                  | -0.08              | 0.00              | -0.01              | 0.05              |

## Compass Verification Results

| Reference ( $^\circ$ ) | Heading Diff. Up ( $^\circ$ ) | Heading Diff. Down ( $^\circ$ ) |
|------------------------|-------------------------------|---------------------------------|
| 0.00                   | 1.02                          | 0.34                            |
| 45.00                  | 0.90                          | 0.19                            |
| 90.00                  | 0.75                          | -0.32                           |
| 135.00                 | 0.41                          | -0.10                           |
| 180.00                 | 0.38                          | 0.12                            |
| 225.00                 | 0.15                          | 0.41                            |
| 270.00                 | 0.48                          | 0.91                            |
| 315.00                 | 0.99                          | 0.84                            |







# Pressure Report

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## Details

|                 |                       |
|-----------------|-----------------------|
| Instrument Type | Signature100          |
| Instrument S/N  | 100767                |
| Pressure Range  | 1500 dBar             |
| Date            | March 14, 2018        |
| Operator        | Asle Martinsen        |
| Location        | Nortek Factory Norway |
| Result          | Passed                |

## Description

Verification is performed in an automated pressure chamber. Fixed-point measurements are collected to verify the sensor.

Criteria of acceptance is  $\pm 0.1\%$  of full scale.

Reference: Telemark Technologies - Vessel Pressure Sensor Inlet.

## Verification Results

| Reference (dBar) | Pressure Diff. (dBar) | Pressure Diff. (% of FS) |
|------------------|-----------------------|--------------------------|
| 34.51            | -0.09                 | -0.01                    |
| 309.29           | 0.26                  | 0.02                     |
| 608.46           | 0.21                  | 0.01                     |
| 910.96           | -0.06                 | 0.00                     |
| 1210.29          | -0.13                 | -0.01                    |
| 1515.14          | -0.19                 | -0.01                    |

